

Docker Compose

What is Docker Compose?

- Docker compose is a tool to run multiple containers together as one application.
- instead of starting each container manually, you write all the settings in one file and start them with a single command.
- you can
 - start, stop, rebuild services
 - check status
 - see logs
 - scale containers up or down

Docker Compose file:

- The configuration is stored in a YAML file (docker-compose.yml or docker-compose.yaml).
- This file describes:
 - which image to use
 - how many containers
 - ports
 - volumes
 - Environment variables.
- In compose, we call them services (not containers).
 - service = A container's configuration (image, ports, volumes, etc)

What is YAML :-

- Full Form: YAML Ain't Markup Language
- A simple text format for storing data in a human-readable way.
- Not a programming language - only used for configuration

- stores data as
• key-value pairs:

name: rupa

- List (sequences):

fruits:

- apple

- banana

- Rules:

- Use spaces (no tabs)
- indentation is important
- - (hyphen) is used for list items.

Docker Compose File versions,

- versions: 1, 2, 2.x, 3.x
- Different versions add new features.

Basic commands:

<u>Command</u>	<u>Purpose</u>
• <code>docker-compose -f file.yaml up</code>	create & start all services
• <code>docker-compose up -d</code>	start in background
• <code>docker-compose up service1</code>	start a specific service
• <code>docker-compose ps</code>	List running services
• <code>docker-compose down</code>	stop and remove all containers in compose
• <code>docker-compose images</code>	List images used by compose
• <code>docker-compose build servicename</code>	Build / rebuild a service
• <code>docker-compose logs -t servicename</code>	view logs (follow mode)
• <code>docker-compose stop servicename</code>	stop a service.

<u>command</u>	<u>purpose</u>
• <code>docker-compose rm service name</code>	Remove a service's container
• <code>docker-compose exec service cmd</code>	Run command inside a container
• <code>docker-compose exec -d service cmd</code>	Run command in background inside container
• <code>docker-compose up -d --scale service=num</code>	Scale service to multiple containers.

Ansible :-

Introduction :-

- DevOps involve several steps: coding, building, testing, packaging, releasing, configuring, and monitoring.
- Configuration management is an important part of DevOps because:
 - it automates repetitive setup tasks.
 - makes systems more reliable
 - speeds up deployments.
 - keeps environments consistent across all stages (dev, test, prod)

What is configuration management

- Process of configuration and maintaining all environments where software runs.
- includes setting up servers, networks, OS, application & configuration files.
- Ensures all changes are tracked so they can be audited and rolled back if needed.
- Bad configuration management can cause: -
 - system outages
 - Data breaches
 - incomplete or faulty testing.